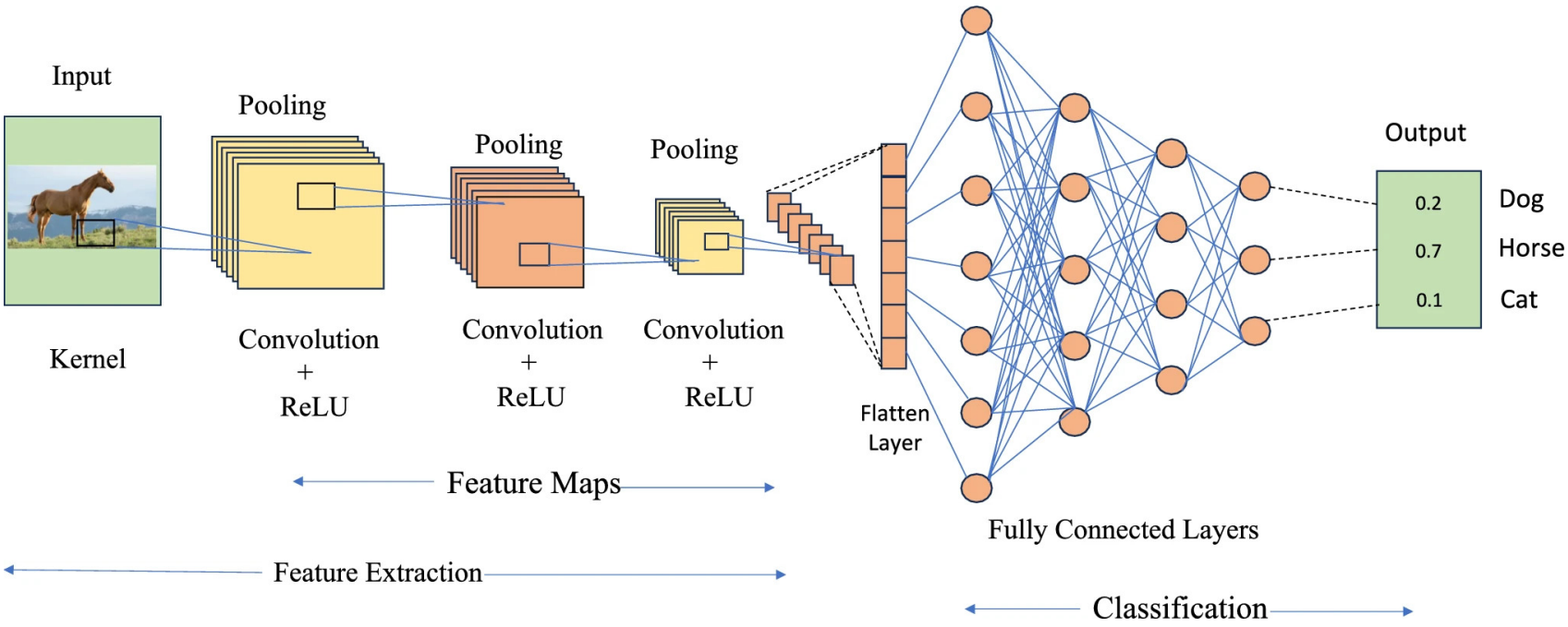
Deep learning for phylogenetics? Learning from classical algorithms

Lena Collienne



Deep Learning

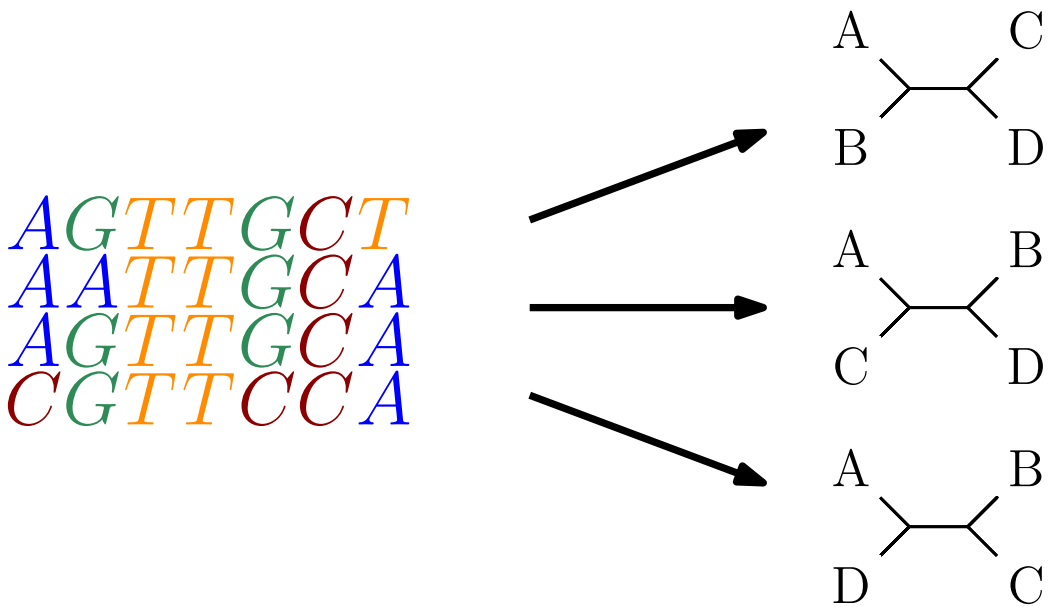


How can we use benefits of deep learning for phylogenetics?

Reza et al. (2026)

First Approaches for Phylogenetic Inference

Classification via CNNs



50 sequences $\rightarrow 2.84 \times 10^{76}$ trees!

Suvorov et al. (2020), Zou et al. (2020), Wang et al. (2023)

First Approaches for Phylogenetic Inference

Estimating distance matrices

Phyloformer (Nesterenko et al. (2025)):

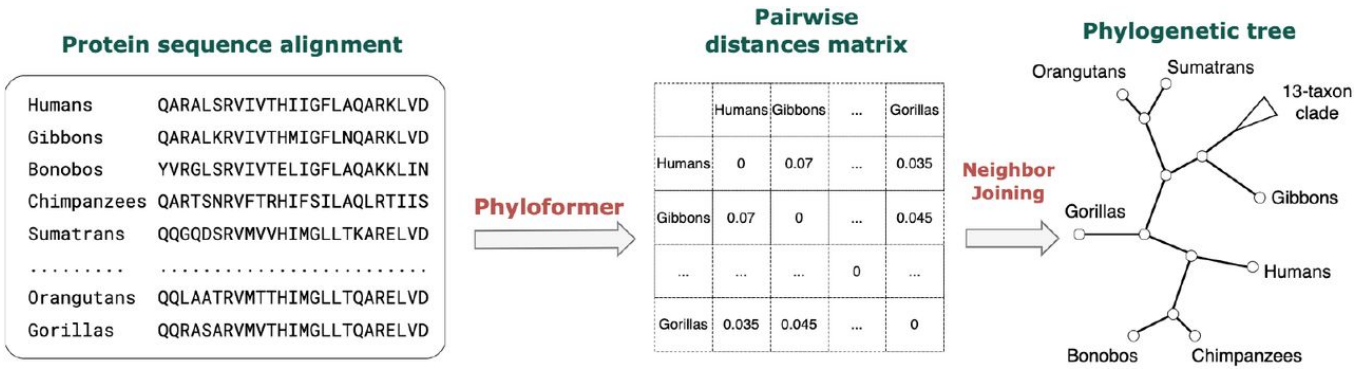


Figure from Ly-Trong et al. (2024)

First Approaches for Phylogenetic Inference

Estimating distance matrices

TreeFormer (Ly-Trong et al. (2024)):

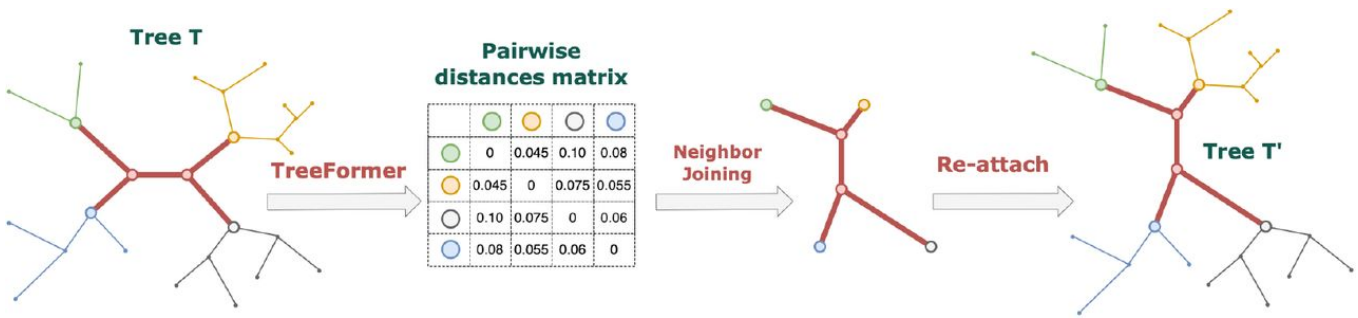


Figure from Ly-Trong et al. (2024)

Designing Deep Learning Methods for Phylogenetics

What we want:

- Trees of varying sizes
- Empirical training data
- Accuracy comparable to existing methods

DPVT

Deep (Neural Networks for) Phylogenetics Via Traversal

Fred Hutch Cancer Center



Harry Richman



David Rich



Chris Jennings-Shaffer



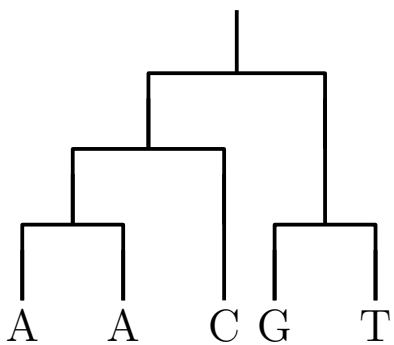
Mary Barker



Erick Matsen

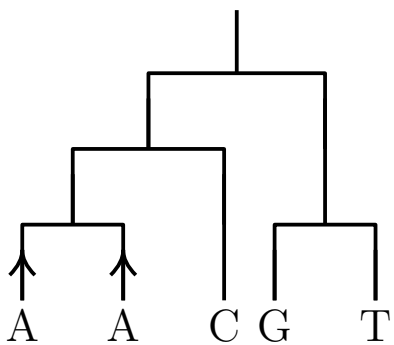
Learning from Classical Algorithms

Dynamic
programming



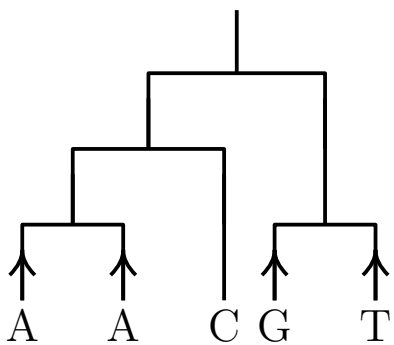
Learning from Classical Algorithms

Dynamic
programming



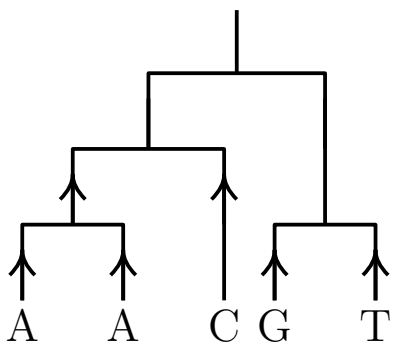
Learning from Classical Algorithms

Dynamic
programming



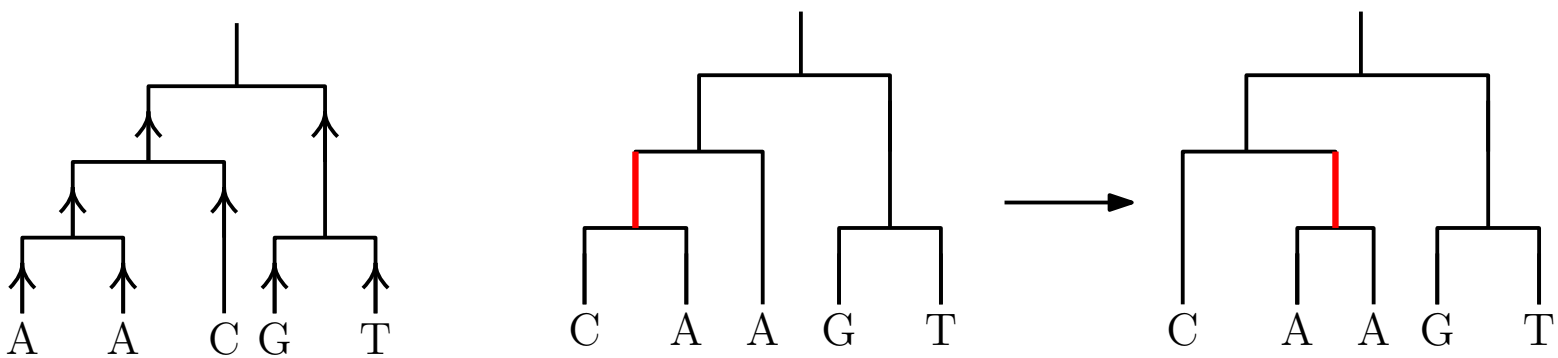
Learning from Classical Algorithms

Dynamic
programming



Learning from Classical Algorithms

Dynamic programming & local search



Learning from Classical Algorithms

Classify Edges - Maximum Parsimony

site i

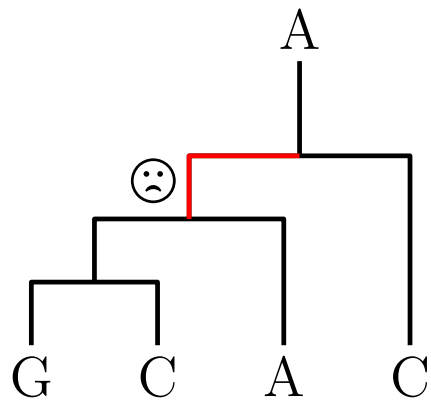
$$
\begin{array}{l}
\ldots \mathrm{G}\,\mathrm{A}\,\boxed{\mathrm{A}}\,\mathrm{C}\ldots \\
\ldots \mathrm{T}\,\mathrm{A}\,\boxed{\mathrm{A}}\,\mathrm{C}\ldots \\
\ldots \mathrm{G}\,\mathrm{A}\,\boxed{\mathrm{C}}\,\mathrm{C}\ldots \\
\ldots \mathrm{T}\,\mathrm{A}\,\boxed{\mathrm{C}}\,\mathrm{C}\ldots \\
\ldots \mathrm{T}\,\mathrm{C}\,\boxed{\mathrm{G}}\,\mathrm{C}\ldots
\end{array}
$$

Learning from Classical Algorithms

Classify Edges - Maximum Parsimony

site i

...GA	A	C...
...TA	A	C...
...GA	C	C...
...TA	C	C...
...TC	G	C...

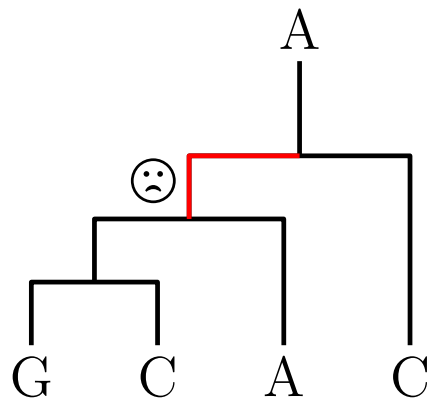


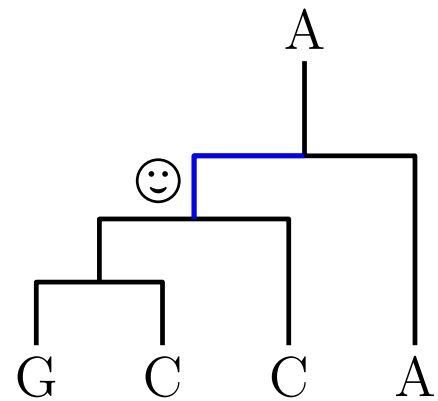
Learning from Classical Algorithms

Classify Edges - Maximum Parsimony

site i

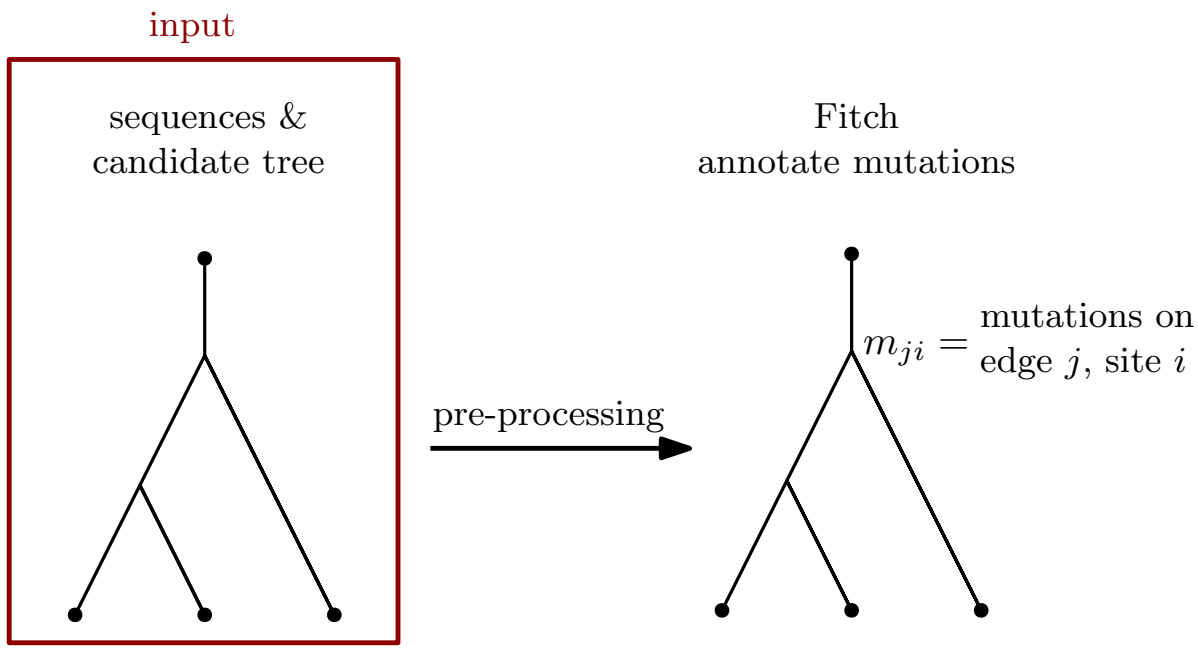
...GA	A	C...
...TA	A	C...
...GA	C	C...
...TA	C	C...
...TC	G	C...





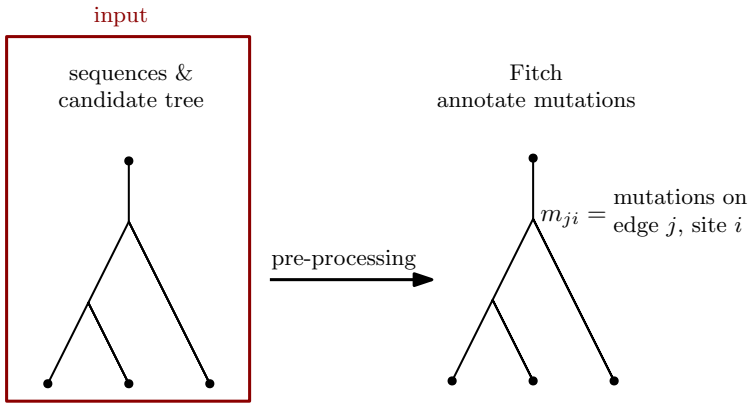
Learning from Classical Algorithms

Pre-processing



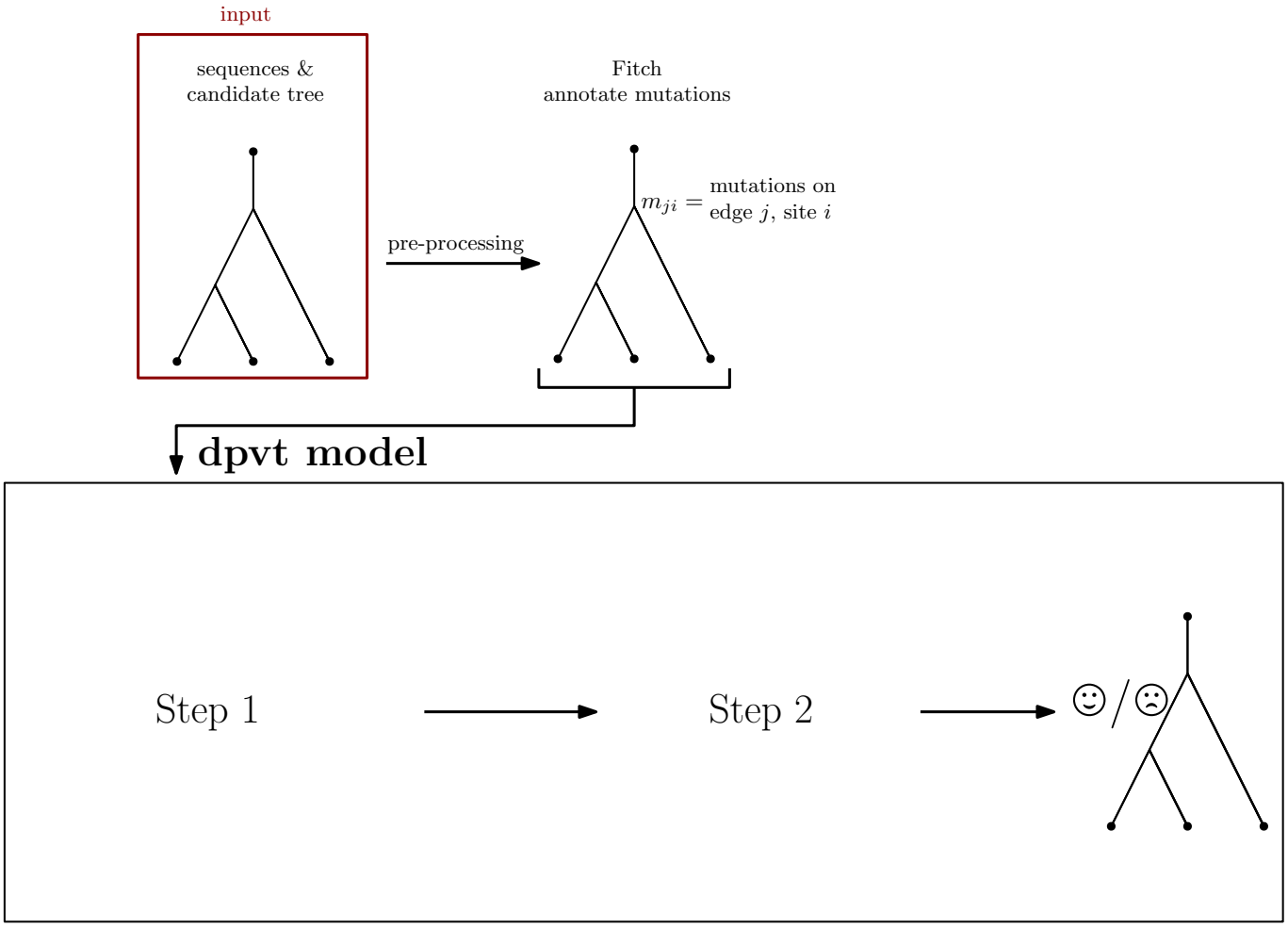
Learning from Classical Algorithms

Overview



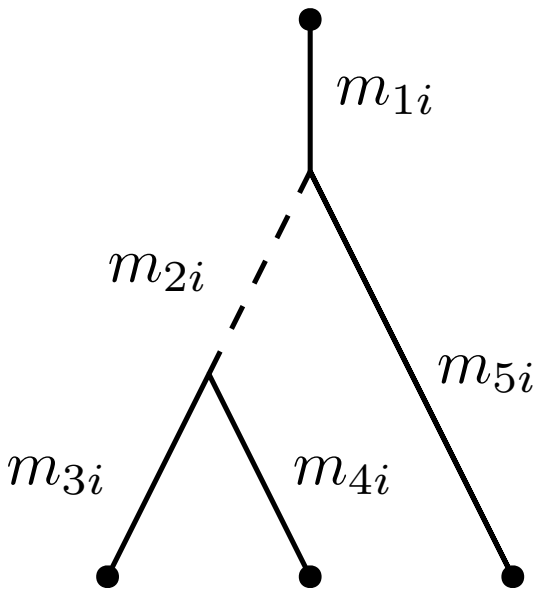
Learning from Classical Algorithms

Overview



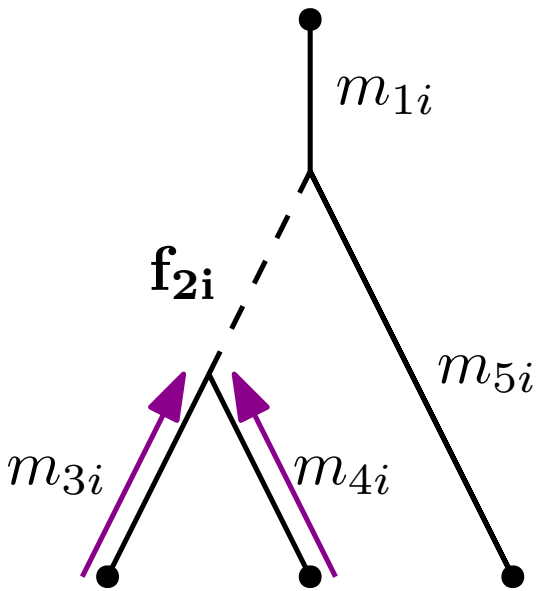
Learning from Classical Algorithms

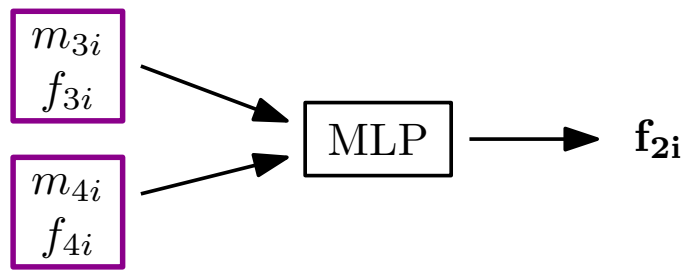
Tree Traversal - per site i



Learning from Classical Algorithms

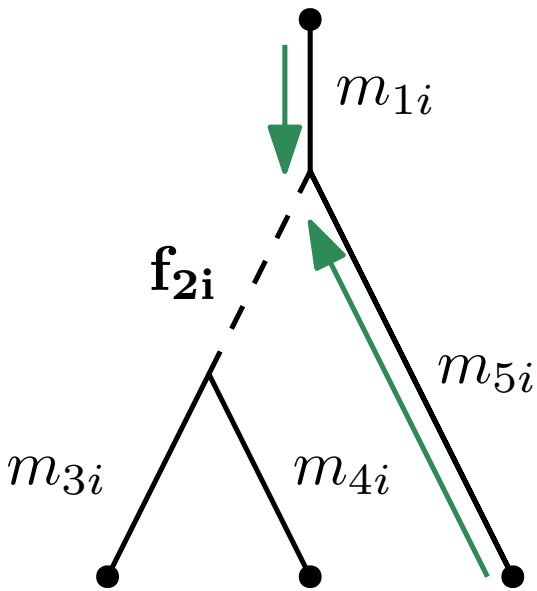
Tree Traversal - per site i

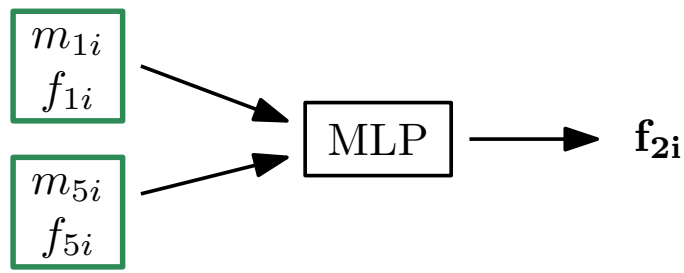




Learning from Classical Algorithms

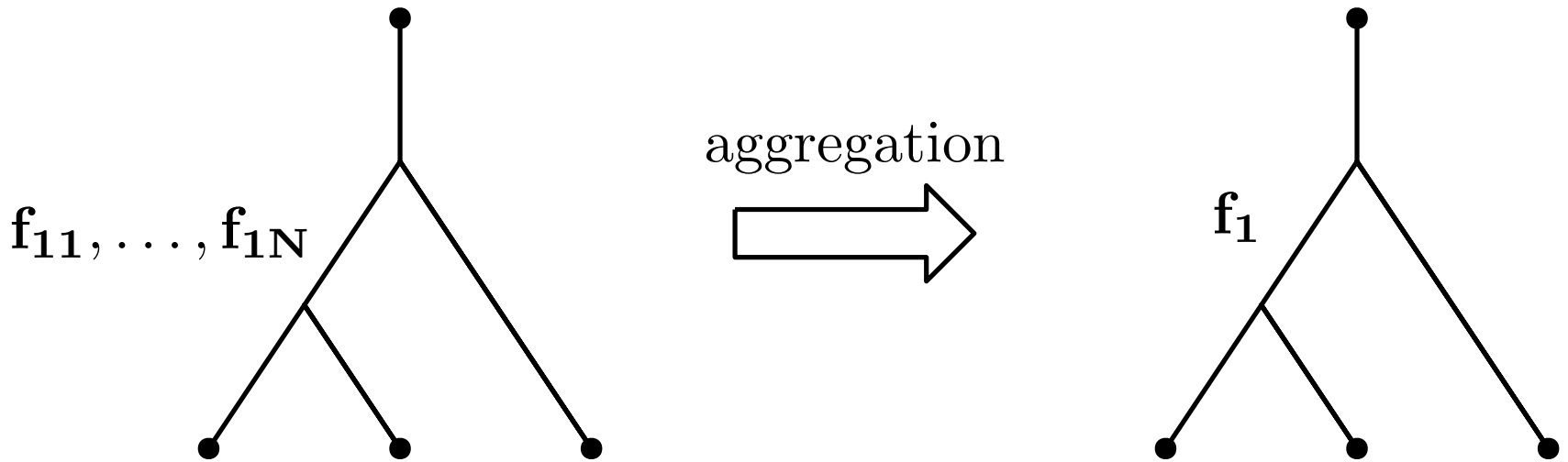
Tree Traversal - per site i





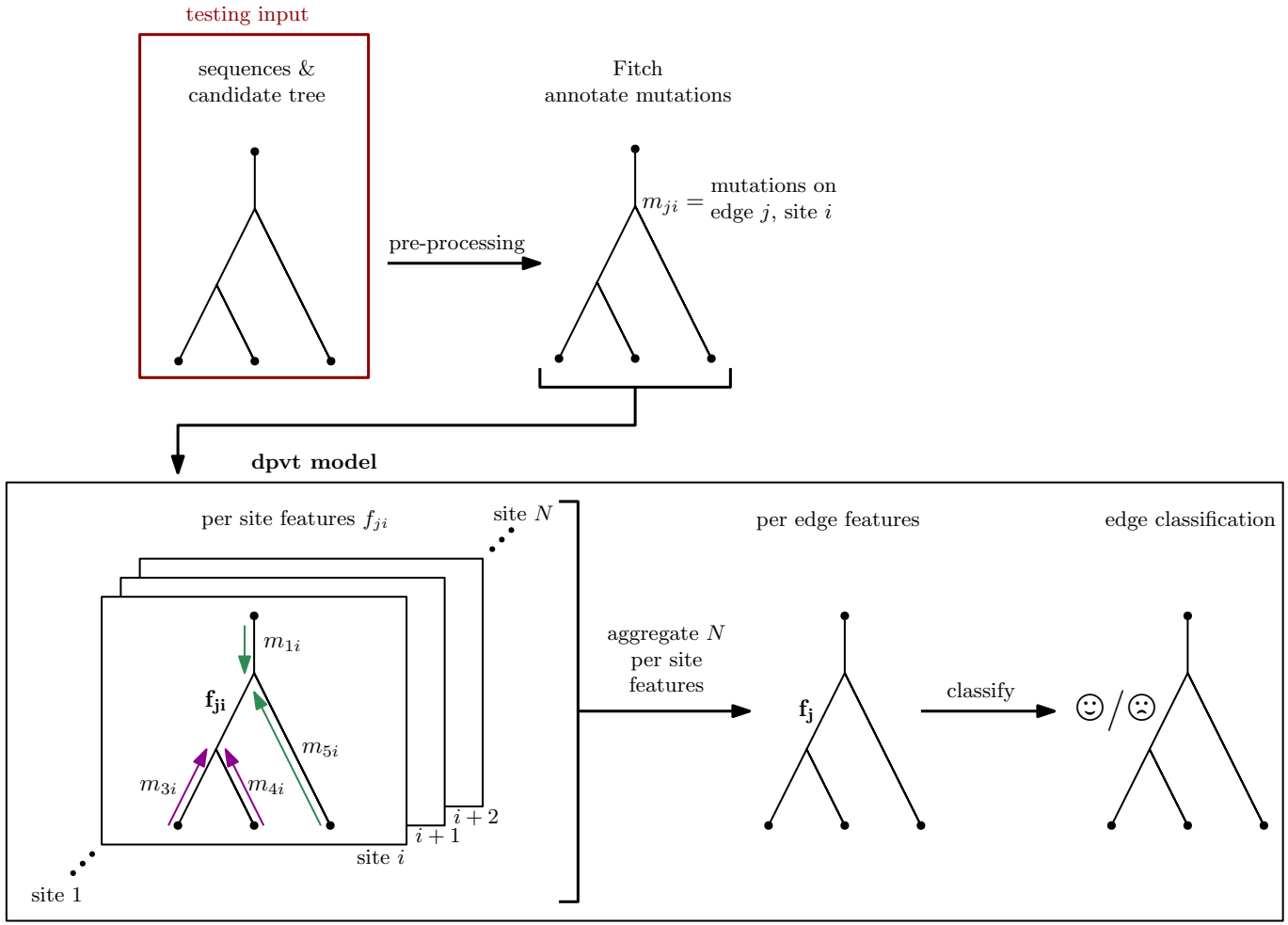
Learning from Classical Algorithms

Aggregation over Sites

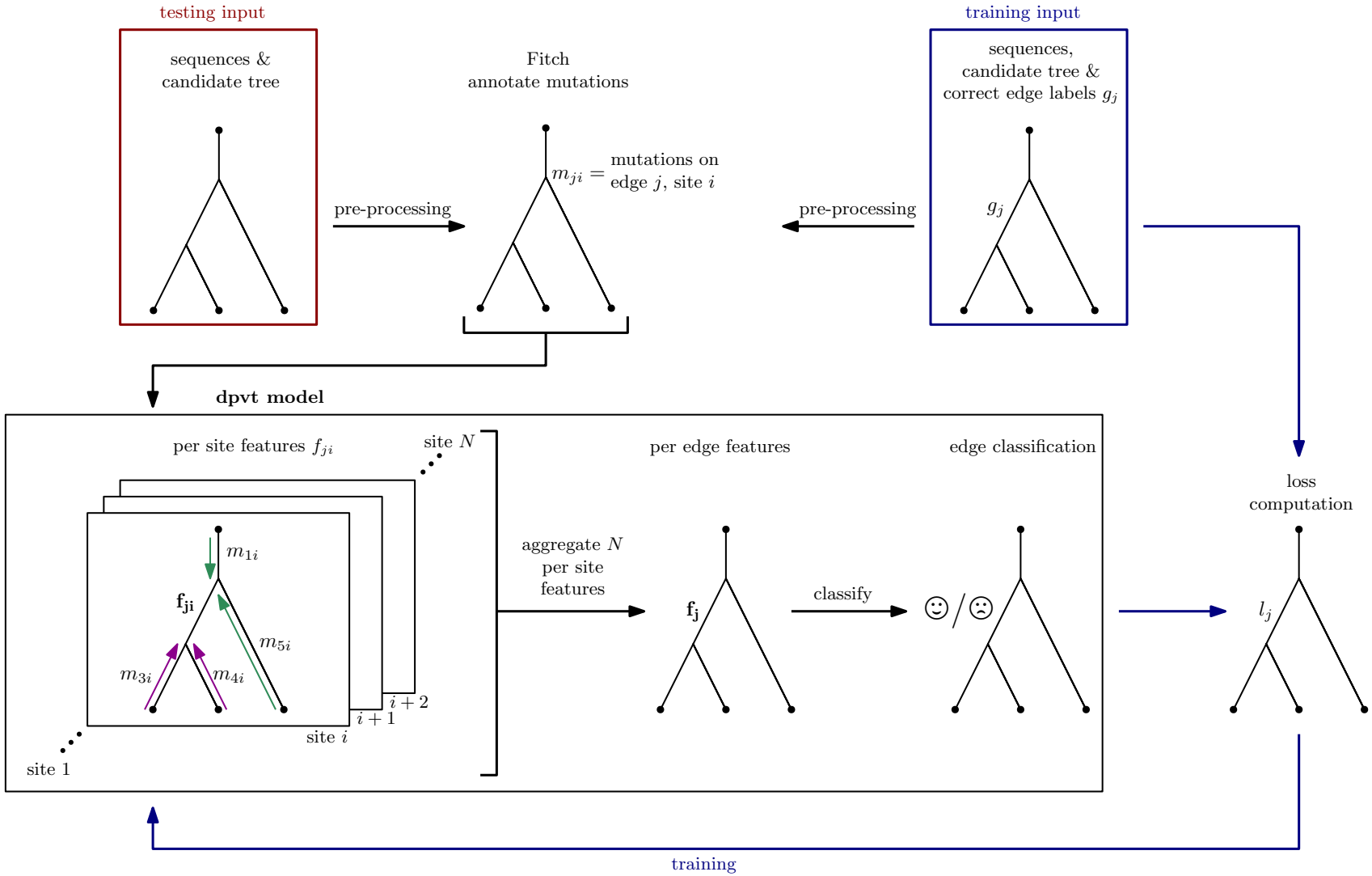


- Average
- Maximum

DPVT


testing input
sequences &
candidate tree
Fitch
annotate mutations
m_{ji} = mutations on edge j, site i
pre-processing
dpvt model
per site features f_{ji}
site N
per edge features
edge classification
m_{1i}
$\mathbf{f_{ji}}$
m_{5i}
m_{3i}
m_{4i}
$i+2$
$i+1$
site i
site 1
aggregate N per site features
$\mathbf{f_j}$
classify

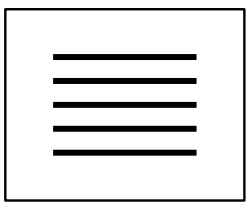
DPVT


testing input
sequences &
candidate tree
Fitch
annotate mutations
m_{ji} = mutations on edge j, site i
pre-processing
pre-processing
training input
sequences,
candidate tree &
correct edge labels g_j
g_j
dpvt model
per site features f_{ji}
site N
per edge features
edge classification
loss
computation
m_{1i}
$\mathbf{f_{ji}}$
m_{5i}
m_{3i}
m_{4i}
$i+2$
$i+1$
site i
site 1
aggregate N
per site
features
$\mathbf{f_j}$
classify
l_j
training

Data Generation

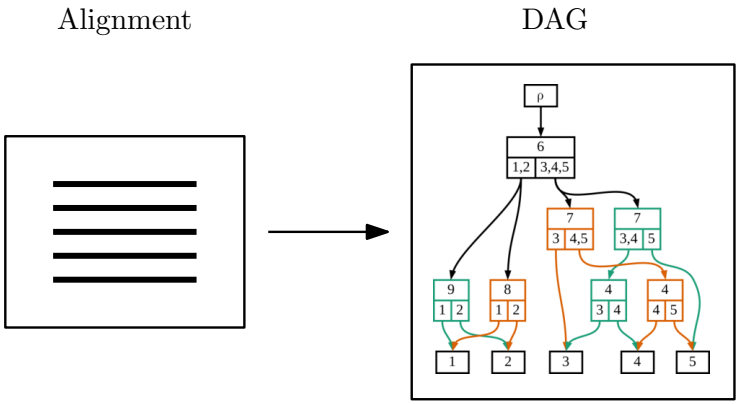
with Larch (Barker et al. (2025))

Alignment



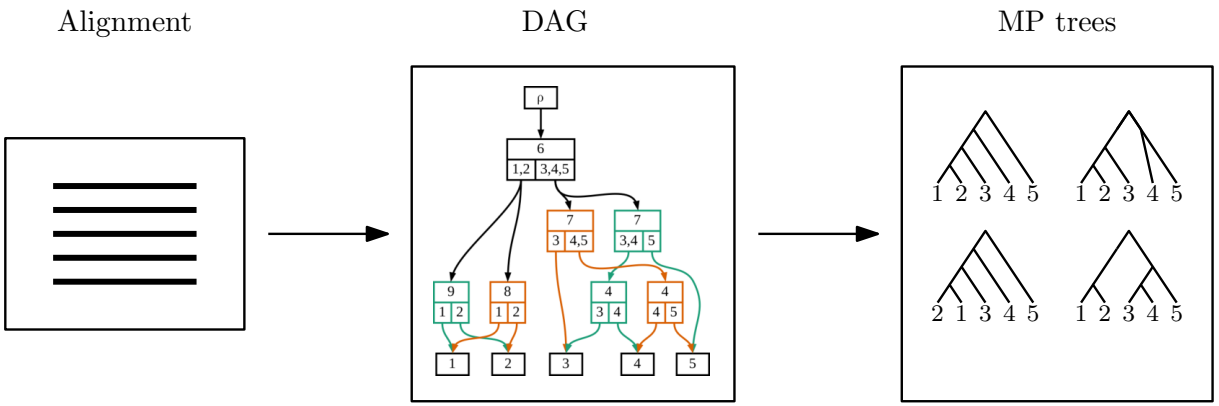
Data Generation

with Larch (Barker et al. (2025))



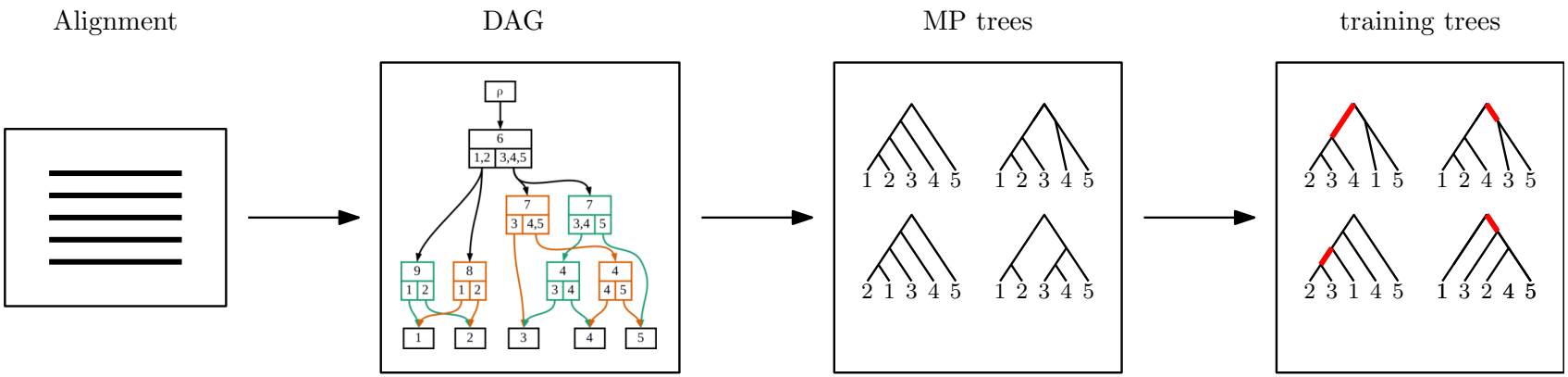
Data Generation

with Larch (Barker et al. (2025))

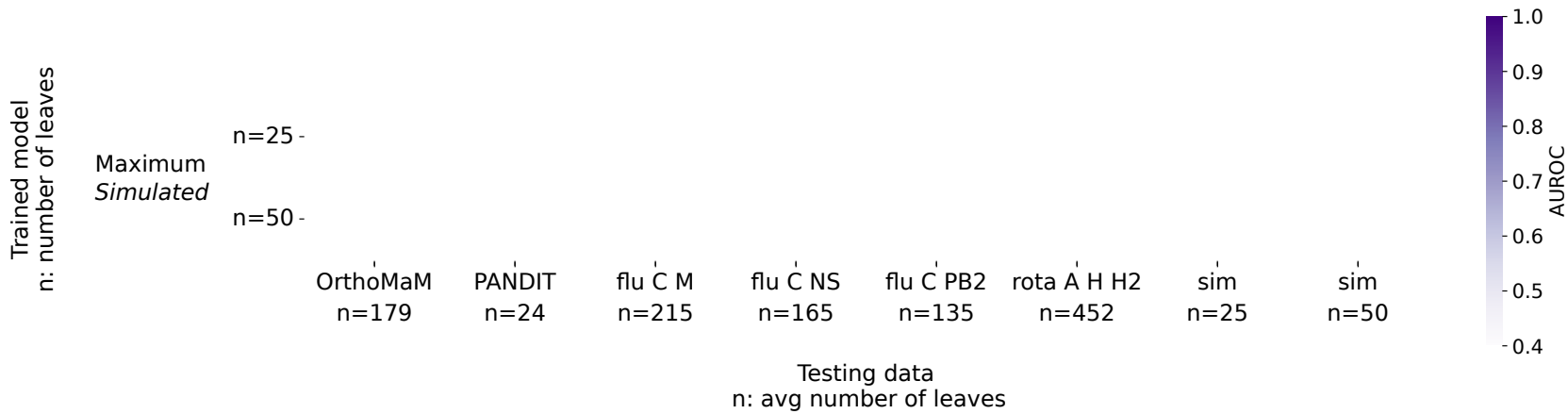


Data Generation

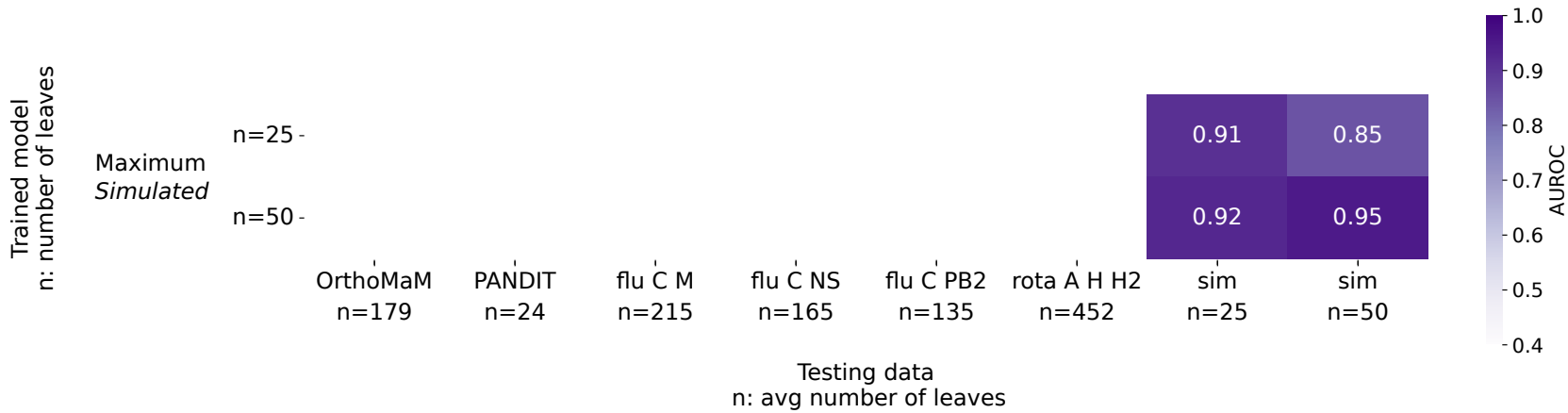
with Larch (Barker et al. (2025))



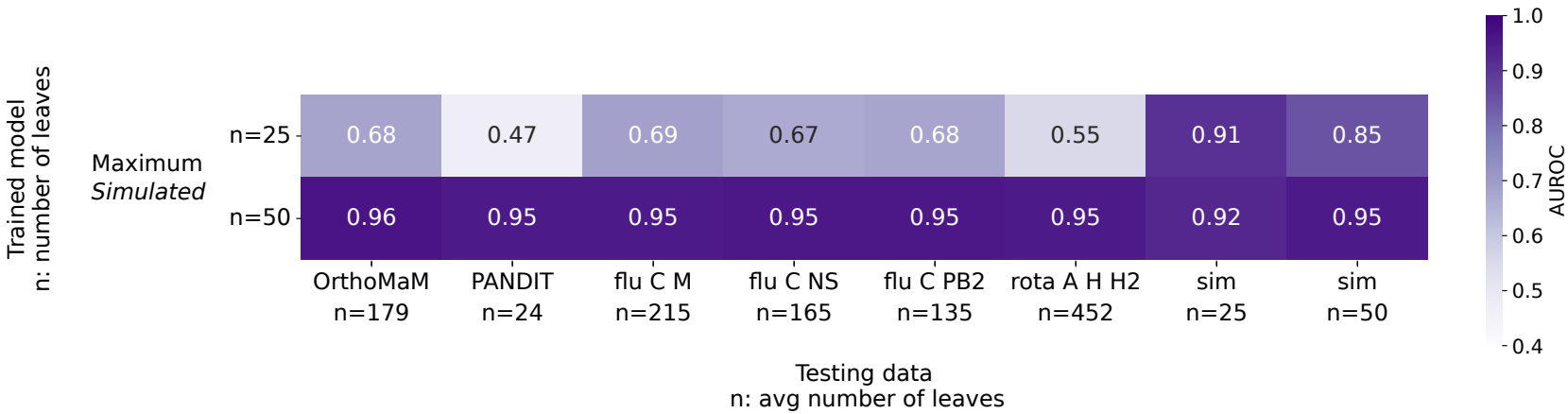
Does DPVT work?



Does DPVT work?



Does DPVT work?

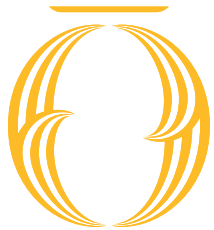


What's Next?

- Shallowfication
- Tree search informed by DPVT
- Tree proposals for MCMC
- Multiple Sequence Alignments ?

Thank You





University of Otago
ŌTĀKOU WHAKAIHU WAKA



Fred Hutch
Cancer Center

hhmi
Howard Hughes
Medical Institute